

PROJECT REPORT

Customer Satisfaction Prediction

1. Introduction

In today's customer-centric business environment, understanding customer satisfaction is essential for improving service quality and maintaining customer loyalty. Organizations collect large volumes of customer support data, which can be analyzed to identify trends, issues, and customer behavior.

This project focuses on analyzing customer support ticket data and extracting meaningful insights using data visualization and machine learning techniques.

2. Objective of the Project

The main objectives are:

- To analyze customer support ticket data
 - To identify common customer issues
 - To understand customer satisfaction patterns
 - To visualize trends and customer behavior
 - To support business decision-making using data
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3. Dataset Description

The dataset contains customer support ticket information.

Key Features:

Feature	Description
Ticket Subject	Issue raised by customer
Ticket Type	Category of issue
Ticket Status	Status (Open, Closed, Pending)
Ticket Priority	Priority level
Customer Satisfaction Rating	Rating (1–5)
Customer Age	Age of customer
Customer Gender	Gender
Product Purchased	Product name
Ticket Channel	Source (Email, Chat, Call)
Date of Purchase	Date

4. Project Workflow Overview

The project follows these steps:

1. Data Loading
2. Data Cleaning

3. Exploratory Data Analysis (EDA)
4. Trend Analysis
5. Customer Segmentation
6. Visualization
7. Dashboard Creation

5. Data Preprocessing

Steps performed:

- Converted date column into datetime format
- Created new feature: **Year Month**
- Handled missing values
- Removed infinite values

Why this is important:

- Ensures data consistency
- Enables time-based analysis
- Prevents errors in visualization

6. Exploratory Data Analysis (EDA)

EDA is used to understand patterns and relationships in data.

6.1 Top 10 Common Issues

Why used:

- To identify most frequent customer problems

Insight:

- Helps companies focus on major issues

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Top 10 Common Issues:
Ticket Subject
Refund request      576
Software bug        574
Product compatibility 567
Delivery problem    561
Hardware issue       547
Battery life         542
Network problem      539
Installation support 530
Product setup        529
Payment issue        526
Name: count, dtype: int64
```

6.2 Ticket Trends Over Time (Line Chart)

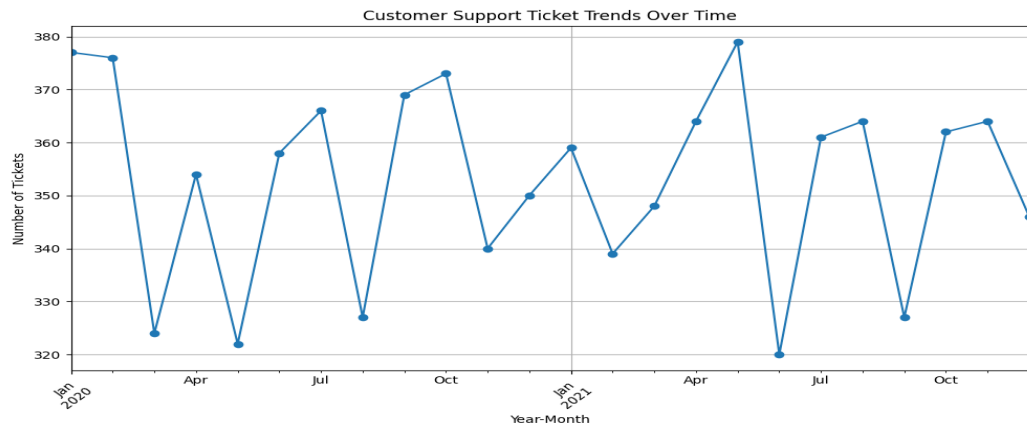
Why used:

- To analyze trends across time

Insight:

- Detects:
 - Increase in complaints

- Seasonal patterns
- Service issues



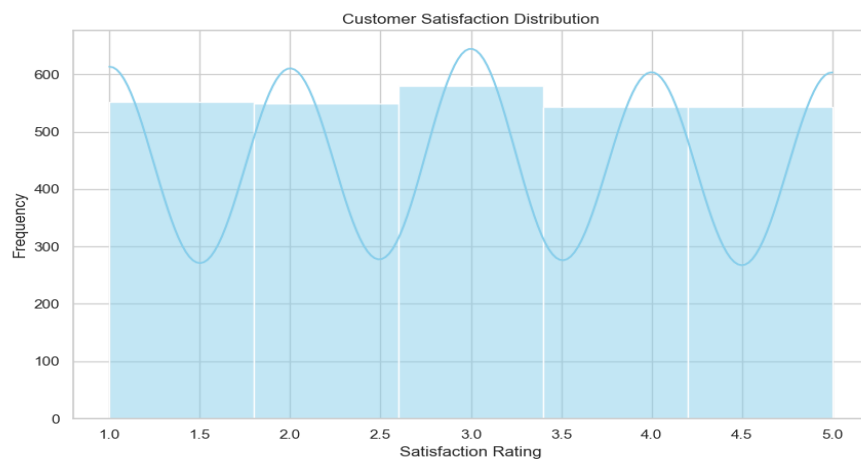
6.3 Customer Satisfaction Distribution (Histogram)

Why used:

- To see how satisfaction ratings are distributed

Insight:

- Shows whether customers are mostly happy or dissatisfied



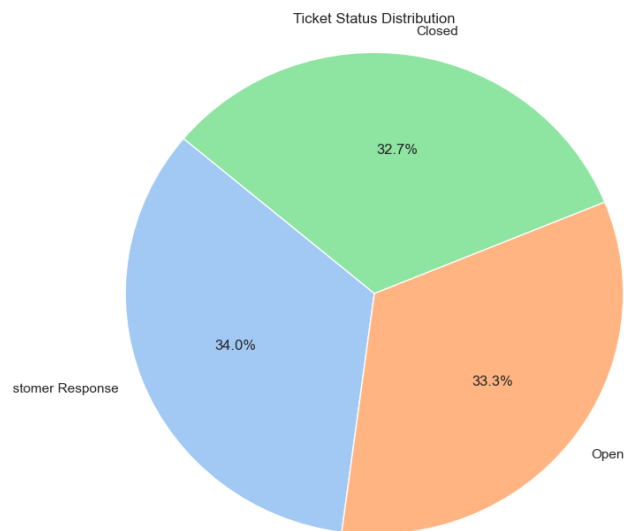
6.4 Ticket Status Distribution (Pie Chart)

Why used:

- To show proportion of ticket statuses

Insight:

- % of resolved vs pending tickets



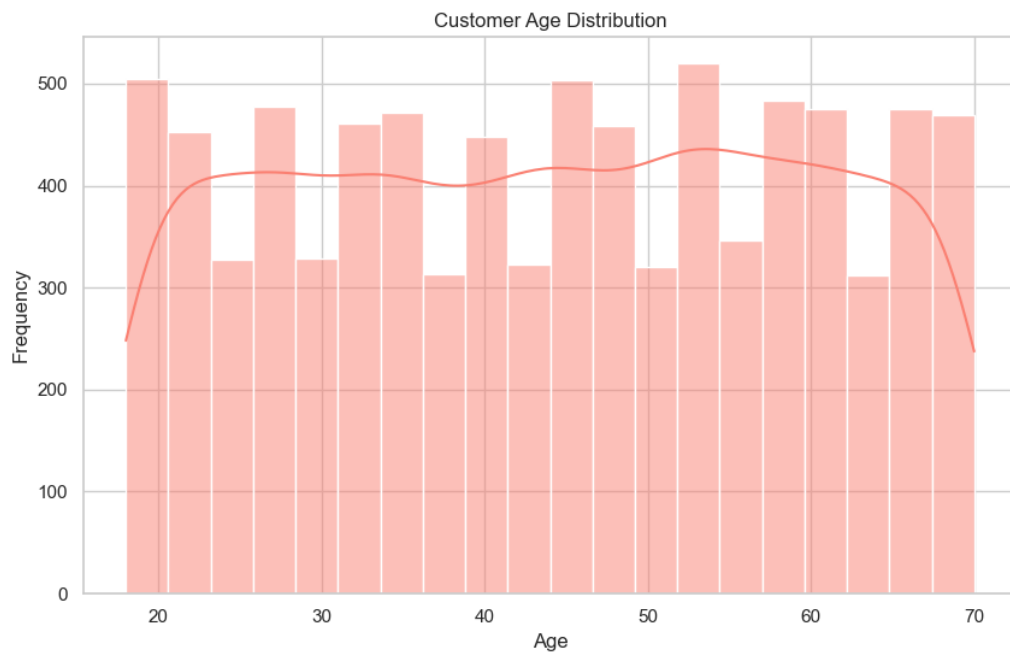
6.5 Customer Age Distribution (Histogram)

Why used:

- To understand age demographics

Insight:

- Identifies dominant customer age groups



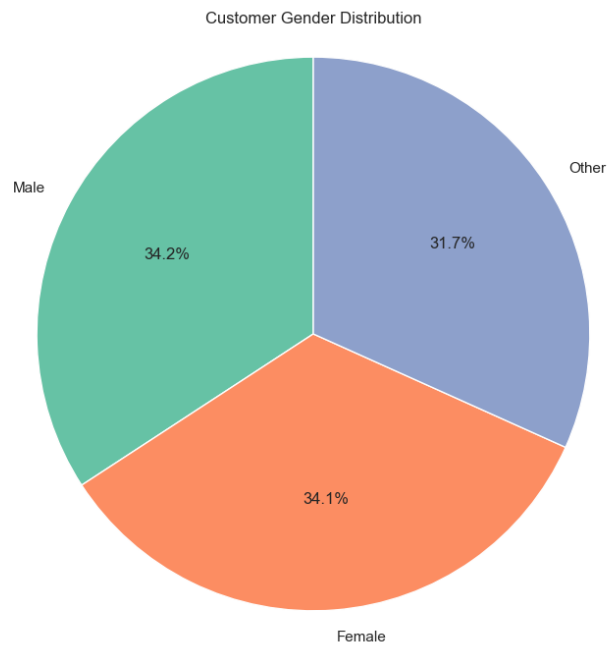
6.6 Customer Gender Distribution (Pie Chart)

Why used:

- To analyze gender-based distribution

Insight:

- Helps in demographic targeting



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6.7 Ticket Channel Distribution (Bar Plot)

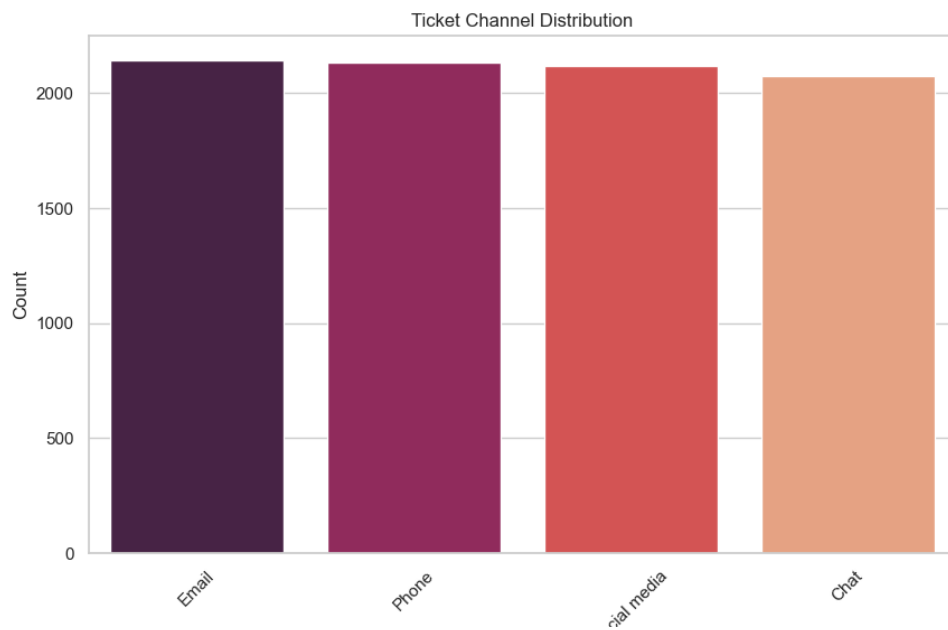
Why used:

- To compare different categories

Insight:

- Identifies most used support channel

📌 Bar plot is best for **categorical comparison**



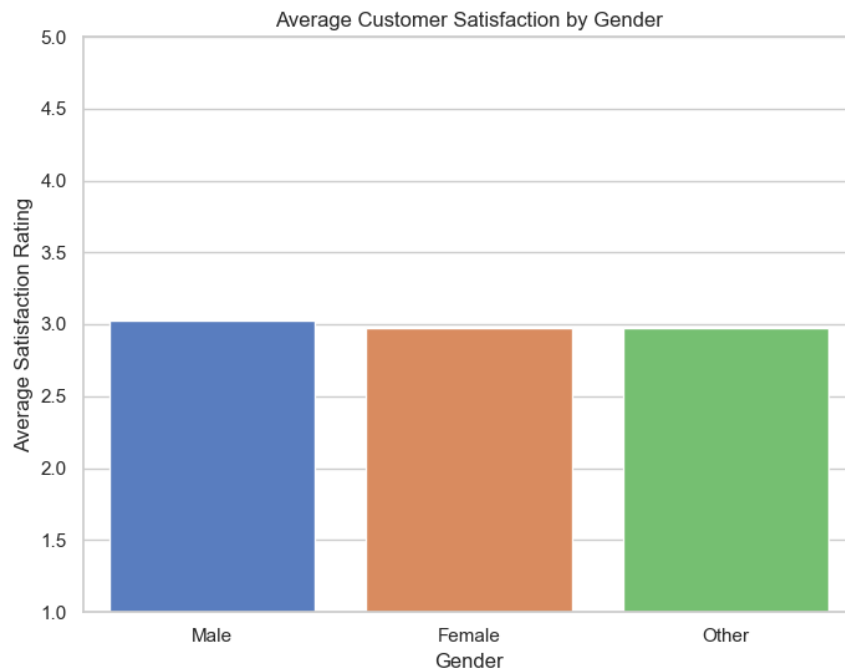
6.8 Satisfaction by Gender (Bar Plot)

Why used:

- To compare average satisfaction

Insight:

- Detects differences in satisfaction levels



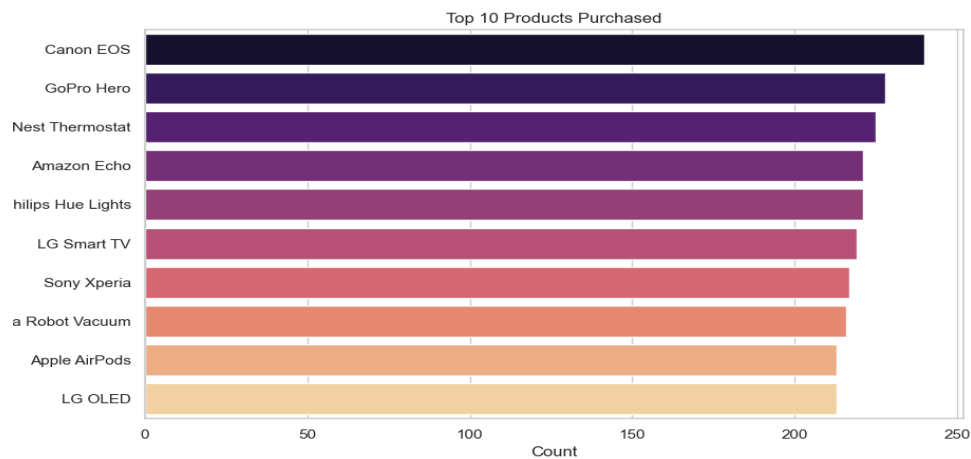
6.9 Product Purchased Analysis

Why used:

- To identify popular products

Insight:

- Helps in marketing decisions



6.10 Ticket Type & Priority (Pie Charts)**Why used:**

- To show distribution of categories

Insight:

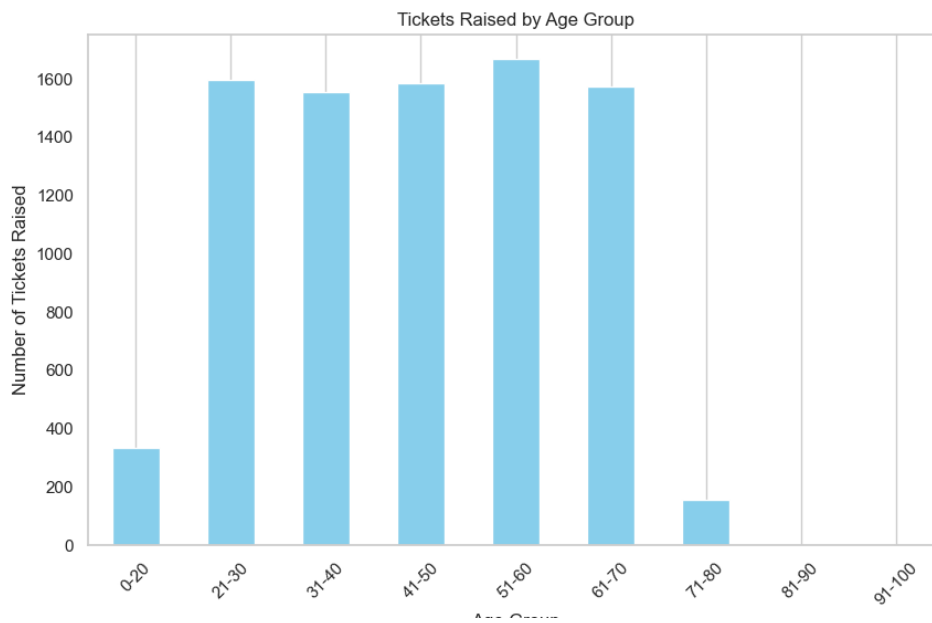
- Helps in workload prioritization

6.11 Age Group Analysis (Bar Chart)**Why used:**

- To group continuous data into categories

Insight:

- Identifies which age group raises more tickets



6.12 Facet Grid (Advanced Visualization)

Why used:

- To compare multiple distributions simultaneously

Insight:

- Shows age distribution across ticket types



7. Dashboard Creation

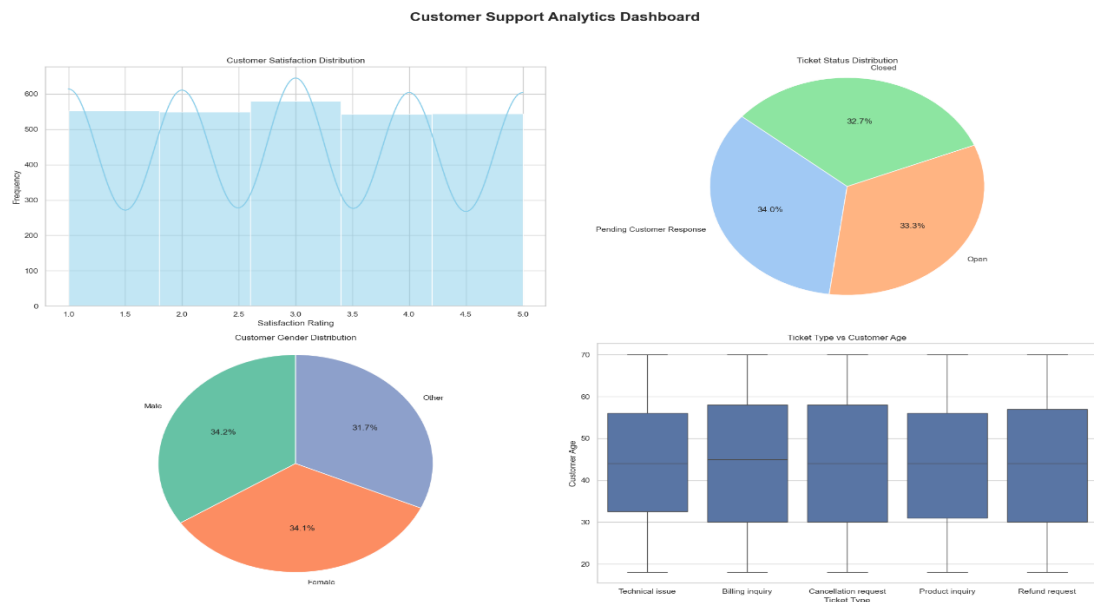
What was done:

A dashboard combining multiple visualizations:

- Satisfaction Distribution
- Ticket Status
- Gender Distribution
- Ticket Type vs Age

Why important:

- Provides a **single-view summary**
- Helps stakeholders make quick decisions



8. Machine Learning Components

Although the project focuses more on analysis, some ML tools were imported:

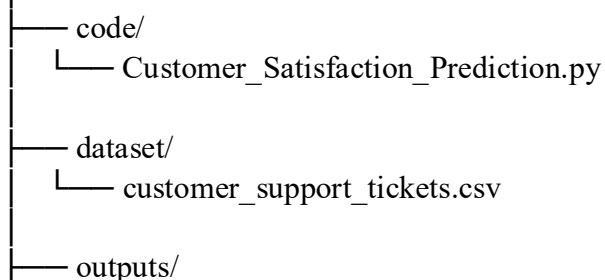
- Linear Regression → for prediction
- KMeans → for clustering

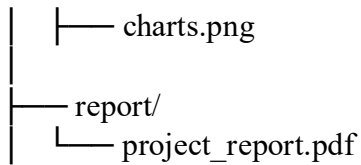
Purpose:

- Predict customer satisfaction
- Segment customers

9. Project Structure

Customer_Satisfaction_Project/





10. Tools & Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

11. Future Scope

- Build predictive models for satisfaction
- Use NLP for ticket text analysis
- Real-time dashboard deployment
- Integration with business systems

12. Key Insights

- Certain ticket types occur frequently
- Customer satisfaction varies across groups
- Some channels are used more than others
- Age and gender influence behavior

13. Conclusion

This project demonstrates how customer support data can be transformed into meaningful insights using data analysis and visualization. It helps organizations improve customer experience by identifying patterns and problem areas.